

Functional Test Cases: Use Case Testing

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed from a black-box, user-centric perspective, based on the key use cases defined in the SRS (Software Requirements Specification). Actual results and status are to be filled during test execution. Each test case is designed to cover the main flow and specified alternative/exception flows.

I.D Functional Test Cases: Use Case Testing

Table 1: Use Case Test Cases

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
Sub-Section: UC-002: Add New Book (Librarian)			
TC_UC_BOOK_001	Successful addition of a new book (Main Flow)	- As a logged-in Librarian, navigate to the Book Management page ('/books'). - Click the "New Book" button. - Fill in all required fields (Name, ISBN) with valid data. - Select a valid Author and Genre from the provided lists. - Ensure "Is Available" is set to true. - Submit the form.	- The system creates a new book record via 'POST /api/book/add'. - A success message is displayed (e.g., "Book added"). - The "Add Book" form closes. - The new book appears in the list on the Book Management page.
TC_UC_BOOK_002	Attempt to add a book with missing required fields (Alternative Flow A1)	- As a logged-in Librarian, navigate to the "Add Book" form. - Fill in the form but leave a required field (e.g., Name) blank. - Submit the form.	- The system rejects the submission due to a validation error. - An error message is displayed, indicating the required field is missing. - The form remains open for correction. - The book is not added to the database.
TC_UC_BOOK_003	Attempt to add a book that fails backend validation (Alternative Flow A2)	- As a logged-in Librarian, navigate to the "Add Book" form. - Fill in all fields with data that is syntactically valid on the client-side but might trigger a server error (e.g., a duplicate ISBN if the server enforces uniqueness). - Submit the form.	- The system displays a generic error message from the API (e.g., "Something went wrong, please try again"). - The book is not added to the database.
Sub-Section: UC-003: Borrow a Book (Member)			
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Table 1 – continued from previous page

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
TC_UC_BORROW_001	Successful borrowing of an available book (Main Flow)	- As a logged-in Member, navigate to the Borrowal Management page ('/borrowals'). - Click the "New Borrowal" button. - Select a book that is marked as available from the list. - Verify that the Member field is correctly auto-populated. - Submit the form.	- A new borrowal record is created with "Borrowed" status via 'POST /api/borrowal/add'. - The availability of the book is updated to 'false'. - A success message is displayed. - The member's borrowal list is updated to show the new borrowal.
TC_UC_BORROW_002	Attempt to borrow an unavailable book (Alternative Flow A1)	- As a logged-in Member, navigate to the "Add Borrowal" form. - Attempt to select a book that is not available (isAvailable=false).	- The UI should ideally prevent the selection of an unavailable book. - If selection is possible, the system should prevent submission or display an error message (e.g., "Book is not available") upon submission attempt. - No borrowal record is created.
TC_UC_BORROW_003	Verify correct data population in borrowal form	- As a logged-in Member, open the "Add Borrowal" form. - Observe the date fields.	- The Member field is correctly auto-populated with the current user's ID. - The Borrowed Date is auto-populated to the current date. - The Due Date is auto-populated to a future date (e.g., 14 days from current). - The initial status is correctly set (e.g., "Borrowed").
Sub-Section: UC-005: View Borrowal History (Member)			
TC_UC_HISTORY_001	View non-empty borrowal history (Main Flow)	- Log in as a Member who has previously borrowed one or more books. - Navigate to the Borrowal Management page ('/borrowals').	- The system fetches all borrowals and the client filters them. - A list/table is displayed showing *only* the records where the 'memberId' matches the logged-in Member's ID. - The displayed data (Book Name, Borrowed Date, Due Date, Status) is accurate for each record.
TC_UC_HISTORY_002	View empty borrowal history (Alternative Flow A1)	- Log in as a new Member with no borrowal records. - Navigate to the Borrowal Management page ('/borrowals').	The system displays a message indicating there is no borrowal history (e.g., "No borrowals found").

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Table 1 – continued from previous page

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
TC_UC_HISTORY_003	Verify role-based view of borrowal history	• Prerequisite: There are borrowals from multiple members in the system. • Log in as a Librarian and navigate to the Borrowal Management page ('/borrowals'). • Log out and log in as a Member (who has at least one borrowal) and navigate to the same page.	• As the Librarian, all borrowal records for all members are visible. • As the Member, only their own borrowal history is visible.
Note: UC-001 (User Login) and UC-004 (Register New User) are detailed in the Authentication & Authorization test plan.			